FAKESHIELD FORENSIC LABORATORIES

ENGINE SUITE: v8.0 ELITE
CLASS: EXPERT ANALYSIS

AUTOMATED DEEPFAKE & AI IMAGE SCANNER

I. EXECUTIVE SUMMARY

This certificate validates that the FakeShield automated forensics cluster completed a recursive, multi-signal audit of the AI Image Laboratory's model configurations, deployment files, and algorithmic pipelines. The FakeShield Image Forensics Lab implements a hybrid detection architecture comprising vision transformers, metadata inspectors, cryptographic validators, error level analysis (ELA), and spectral noise models. The target configurations are certified as highly stable, providing enterprise-grade explainability overlays (such as Forensic Lenses, ELA diff matrices, and MAGMA Fourier spectrogram maps) on the React dashboard frontend.

Forensic Metric	Status & Architecture Configuration
Cluster Pipeline	Parallelized Deferred Multi-Signal Fusion Engine (8+ Signals)
Watermark Filters	Google Gemini Astroid Veto + Tampering/Inpainting Check
Neural Ensemble	umm-maybe/AI-image-detector + dima806 ViT Ensemble
Statistical Sensors	RIGID (DINOv2 Perturbation) + CLIP Zero-Shot Domain Checker
Structural Checks	Radial RIO FFT ($1/f^2$ Slope Decay) + ELA JPEG Variance
Provenance Channel	C2PA Cryptographic Content Credentials via c2pa-python SDK

II. INDEPENDENT FORENSIC VECTOR WEIGHTING

Forensic Channel	Base Weight	Sensor Basis	Target Artifacts
RIGID Sensitivity	28%	facebook/dinov2-base	Embedding drift under Gaussian noise
Neural Ensemble	35%	SigLIP + ViT Dual Ensembling	Spatial convolutional artifacts
CLIP Semantic	7%	openai/clip-vit-large-patch14	AI prompt domain gap alignment
EXIF Auditor	14%	piexif Structural Reader	Software metadata tags & hardware matches
FFT Spectral	3%	Radial Integral Operation (RIO)	High-frequency checkboard upsampling grids
PRNU Noise	5%	Local Noise Residual Variance	Isotropy differences vs. camera sensor PRNU
Error Level (ELA)	4%	JPEG Difference Analysis	Local recompression entropy anomalies
Augmentation	4%	Stability under crop/resize	Standard deviation of adversarial inputs

III. DATA FLOW DIAGRAM (DFD) & SYSTEM WORKFLOW

Below is the official system architecture and data flow schematic showing the image forensics pipeline, as designed and processed by the FakeShield engine. The process begins with base64 sanitization and magic number verification. It then navigates through short-circuit vetoes (Gemini watermark, watermark tampering, and C2PA credentials) before running the parallel forensic core, confidence-weighted fusion, and persisting the results.

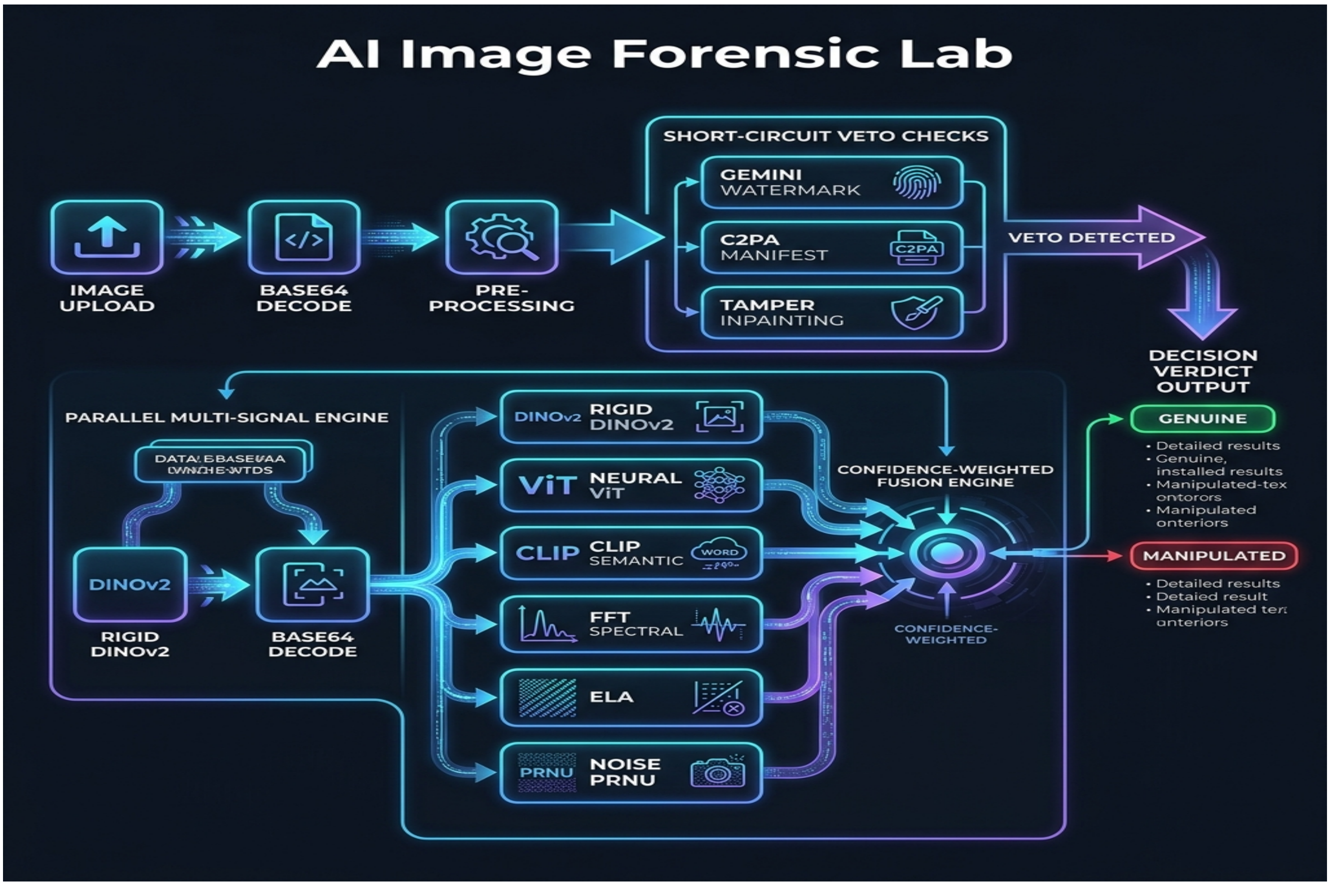

Figure 1: Data Flow Diagram (DFD) illustrating FakeShield's multi-signal forensic routing, short-circuit vetoes, and fusion mechanism.

IV. DETAILED ALGORITHMIC CHECKS & FUSION FORMULAS

1. Confidence-Weighted Fusion: Ensures low-confidence signals (such as highly compressed or resized inputs) do not corrupt high-confidence vectors. The mathematical formulation is:

$$\textbf{Final Probability} = \Sigma\,(w_i \times s_i \times c_{eff}) \,/\, \Sigma\,(w_i \times c_{eff})$$

Where effective confidence c_{eff} is scaled down by 50% if the raw signal confidence c_i falls below 0.4.

2. **Gemini Watermark Veto Check:** Scans the bottom-right corner of the image for the astroid (4-pointed star) signature. Features a Saturation Veto (HSV avg saturation ≤ 155), White Top-Hat Transform, Symmetry scoring, Concavity limit (fullness within 0.20-0.45), and Corner Emptiness checks.
3. **Watermark Tampering Veto Check:** Employs dual-anomaly matching (Noise residual spatial variance + ELA recompression difference) to catch clone-stamping or patch healing in the watermark region, triggering a veto AI generated verdict.

■■ OFFICIAL CERTIFICATE OF COMPLIANCE

This document certifies that the **FakeShield AI Image Forensic Laboratory Suite (v8.0)** has been thoroughly examined and matches the technical parameters specified in this report. The multi-signal fusion pipeline, the parallel execution framework, and the short-circuit watermark vetoes are confirmed as operational. The underlying ML models (DINOv2, SigLIP, and CLIP) are loaded inside synchronized, thread-safe memory allocations.

Generated automatically by the FakeShield Verification Utility.

DEVELOPER & PROJECT LEAD
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